

LANG SON, Viet Nam

WMO: 488300

Lat: 21.8308N

Lon: 106.7690E

Elev: 263

StdP: 98.21

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.1	6.6	-0.7	3.7	11.2	1.3	4.3	10.5	8.9	10.6	8.1	11.0	2.4	0	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.6	34.0	26.4	33.1	26.2	32.3	26.0	27.4	31.9	27.0	31.4	26.7	30.7	2.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	22.4	29.5	25.9	21.9	29.2	25.6	21.5	28.8	89.3	32.2	87.3	31.5	85.7	31.0	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.2	5.2	2.6	35.9	1.6	0.9	1.5	36.6	0.6	37.1	-0.3	37.6	-1.5	38.3		
(5)			2.2	28.5	1.6	0.4	1.1	28.8	0.2	29.0	-0.7	29.2	-1.8	29.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.1	13.8	15.8	18.7	23.1	26.1	27.7	27.8	27.4	26.1	23.5	19.5	15.2		
	DBStd	5.89	4.28	4.89	4.12	3.45	2.49	1.59	1.49	1.45	1.89	2.44	3.25	3.59		
	HDD10.0	19	12	5	1	0	0	0	0	0	0	0	0	2		
	HDD18.3	442	152	99	47	6	0	0	0	0	0	1	24	112		
	CDD10.0	4434	129	168	271	393	499	531	551	538	483	418	286	164		
	CDD18.3	1814	11	29	59	149	241	281	293	280	233	161	60	15		
	CDH23.3	15026	28	122	230	953	2127	2872	2951	2551	1852	990	303	48		
(13)	CDH26.7	4918	1	22	51	257	739	1043	1059	851	592	254	46	3		
(14)	Wind	WSAvg	1.8	2.3	2.2	2.1	1.9	1.7	1.3	1.2	1.5	1.7	2.0	2.2		
(15) Precipitation	PrecAvg	1316	35	32	56	77	163	194	253	215	153	63	46	29		
	PrecMax	1709	135	227	174	167	295	325	534	370	388	166	163	80		
	PrecMin	885	0	1	7	14	69	96	133	78	23	9	1	0		
	PrecStd	222	34	43	44	39	57	60	100	80	80	47	42	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.1	29.2	31.1	33.8	35.1	34.9	34.9	34.3	33.2	31.5	29.6	26.8		
		MCWB	19.9	20.9	21.2	24.6	26.6	26.9	26.9	26.6	25.8	23.9	23.0	21.4		
	2%	DB	23.8	26.8	27.9	31.3	33.3	33.5	33.6	33.2	32.1	30.3	28.0	24.7		
		MCWB	18.8	20.8	22.4	24.7	26.0	26.7	26.6	26.3	25.4	23.5	22.2	19.8		
	5%	DB	22.0	24.6	25.7	29.6	31.9	32.5	32.6	32.2	31.2	29.3	26.6	22.9		
		MCWB	18.0	20.0	21.8	24.2	25.6	26.5	26.3	26.1	25.1	23.1	21.1	18.5		
	10%	DB	20.3	22.7	24.0	28.0	30.6	31.4	31.6	31.0	30.1	28.1	25.1	21.1		
		MCWB	17.4	19.3	21.1	23.4	25.1	26.3	26.1	25.9	24.7	22.7	20.2	17.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	22.2	23.8	26.0	27.5	28.0	27.9	27.7	27.1	25.6	24.2	22.4		
		MCDB	24.1	26.3	27.9	31.0	32.5	32.7	32.7	32.0	31.2	29.4	27.5	25.6		
	2%	WB	19.7	21.4	23.0	25.2	26.8	27.4	27.2	27.1	26.5	24.7	23.4	21.0		
		MCDB	22.5	25.1	26.4	29.7	31.6	31.8	31.7	31.3	30.2	28.2	26.2	23.6		
	5%	WB	18.9	20.7	22.4	24.6	26.2	27.0	26.8	26.6	26.0	24.2	22.6	19.7		
		MCDB	21.2	23.8	25.1	28.5	30.5	31.1	31.0	30.5	29.5	27.5	25.2	21.8		
	10%	WB	18.0	19.9	21.7	24.0	25.7	26.6	26.4	26.2	25.5	23.6	21.8	18.3		
		MCDB	20.0	22.3	23.8	27.4	29.5	30.3	30.2	29.7	28.6	26.7	24.2	20.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	6.4	5.6	6.3	7.0	6.8	6.6	6.6	7.0	7.7	8.2	7.5		
		MCDBR	9.6	9.6	8.4	8.7	9.1	8.6	8.7	8.7	8.7	9.5	10.2	9.9		
		MCWBR	4.7	4.2	3.5	3.2	3.0	2.9	2.7	2.8	2.8	3.4	4.4	4.7		
	5% WB	MCDBR	7.4	8.2	6.8	7.8	7.6	7.5	7.7	7.7	7.5	7.7	8.0	7.6		
		MCWBR	4.1	3.9	3.3	3.3	3.1	3.0	2.8	2.9	2.9	3.1	4.1	4.4		
(40) Clear-Sky Solar Irradiance	taub		0.578	0.609	0.784	0.741	0.550	0.527	0.499	0.562	0.601	0.690	0.575	0.599		
	taud		1.758	1.733	1.446	1.564	2.038	2.120	2.187	1.986	1.851	1.623	1.817	1.731		
	Ebn at Noon		683	696	599	640	770	781	803	757	719	625	681	644		
	Edh at Noon		208	225	312	281	174	159	149	182	205	248	194	207		
(44) All-Sky Solar Radiation	RadAvg		2.12	2.50	2.71	3.52	4.37	4.55	4.65	4.54	4.39	3.77	3.18	2.49		
	RadStd		0.52	0.45	0.36	0.40	0.44	0.46	0.39	0.32	0.29	0.28	0.46	0.44		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	+0.40	+0.30	+0.27	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	+0.60	+0.20	+0.19	N/A	N/A	N/A

Nomenclature: See separate page